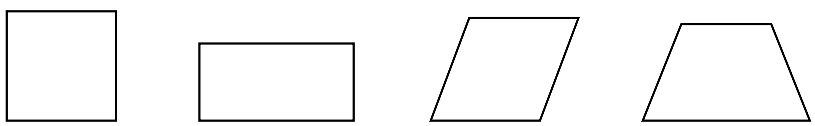
Sorting Four-Sided Shapes by What They Share

Sorting four-sided shapes by square corners, by side lengths, and by parallel sides, and reading a shape-sort graph. Ms. Diaz's class made a set of shape cards. Every card has four straight sides.

- A **square corner** is a corner shaped like the corner of a piece of paper. Use the corner of your paper to test one.
- Two sides are **parallel** when they run the same way and would never bump into each other, like two rails of a train track.
- Shapes are drawn the true size they were cut, so you can compare side lengths by looking and by measuring.

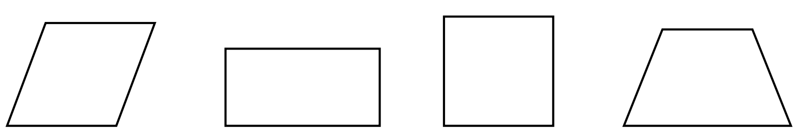
1. Circle every card below that has **four square corners**. Then write how many cards you circled.



A B C D

Answer: ____________________

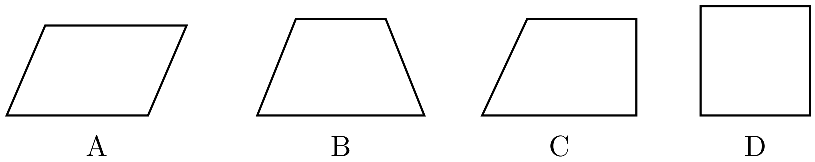
2. Circle every card below whose **four sides are all the same length**. Then write how many cards you circled.



A B C D

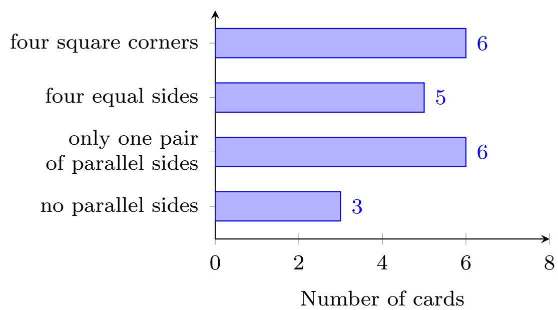
Answer: ____________________

3. A card goes in Ms. Diaz's "one pair" box when it has **exactly one pair of parallel sides**. Circle every card that belongs in the "one pair" box, then write how many cards you circled.



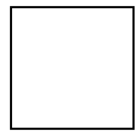
Answer: ________________

4. The class sorted their whole set of cards into four groups and made this graph. **(a)** How many cards did the class sort in all? **(b)** How many more cards went in the "four square corners" group than in the "no parallel sides" group?



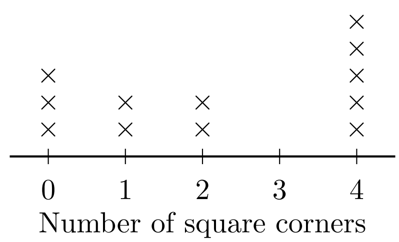
Answer: ________________

5. Rosa says the card below belongs in the rectangle group. Ben says the same card belongs in the square group. Their teacher says they are both right. Use the words *sides* and *corners* to explain how one card can belong in both groups.



Answer: ____________________

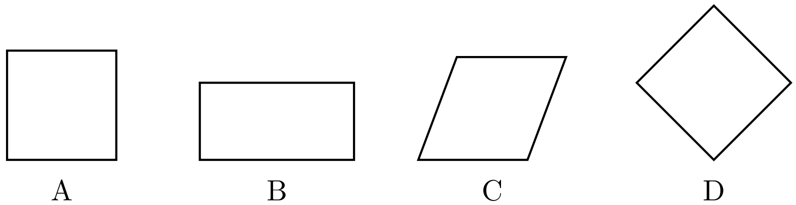
6. Each $\times$ on the line plot stands for one shape card. The number below tells how many square corners that card has.



(a) How many shape cards are on the line plot in all? **(b)** Which number of square corners happened most often? **(c)** No card sits above the mark for three. Explain why a shape with four straight sides can never have exactly three square corners.

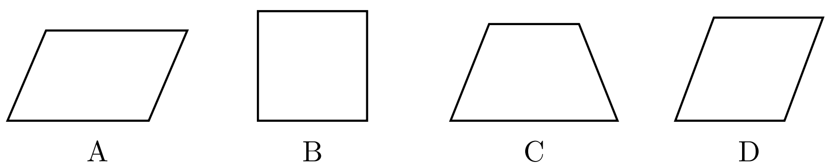
Answer: ____________________

7. Nia made a group and named it "all four sides the same length." Circle every card below that belongs in Nia's group, then write how many cards belong in it.



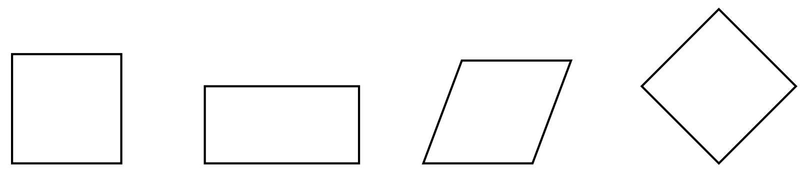
Answer: ________________

8. Kim is filling the "two pairs of parallel sides" box. She looks at the four cards below and says that all 4 of them belong in it. Kim is wrong. Circle the cards that really belong, write how many there are, and write the letter of the card Kim should take back out.



Answer: ________________

9. Some cards belong in *two* groups at once. Circle every card below that has **four square corners** *and* **four sides the same length**, then write how many cards you circled.



A B C D

Answer: ________________

10. On the dot paper below, draw one four-sided shape with *exactly one* pair of parallel sides, and beside it draw one four-sided shape with *two* pairs of parallel sides. Then write one sentence naming something both of your shapes share.